

23AHS21 - probability and Statistics

Assignment - 2 (Unit-2: probability)

- 1) Define i) probability ii) Addition Theorem
 iii) Multiplication Theorem iv) Conditional probability
 v) Random variable vi) Mathematical Expectation.

2) State and prove Baye's Theorem.

3) In a bolt factory machines A, B, C manufacture 20%, 30% and 50% of the total of their output and 6%, 3% and 2% are defective. A bolt is drawn at random and found to be defective. Find the probabilities that it is manufactured from i) Machine A ii) Machine B iii) Machine C.

4) A random variable x has the following probability:

$x :$	0	1	2	3	4	5	6	7
$p(x) :$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

- i) Determine K ii) $p(x < 6)$, $p(x \geq 6)$, $p(0 < x < 5)$, $p(0 \leq x \leq 4)$
 iii) if $p(x \leq K) \geq \frac{1}{2}$, find the minimum value of K
 iv) distribution function of x v) Mean vi) Variance.

5) Suppose a Continuous random Variable x has the probability density

$$f(x) = K(1 - x^2) \quad \text{for } 0 < x < 1,$$

$$= 0 \quad \text{otherwise.}$$

Find i) K ii) Mean iii) Variance.